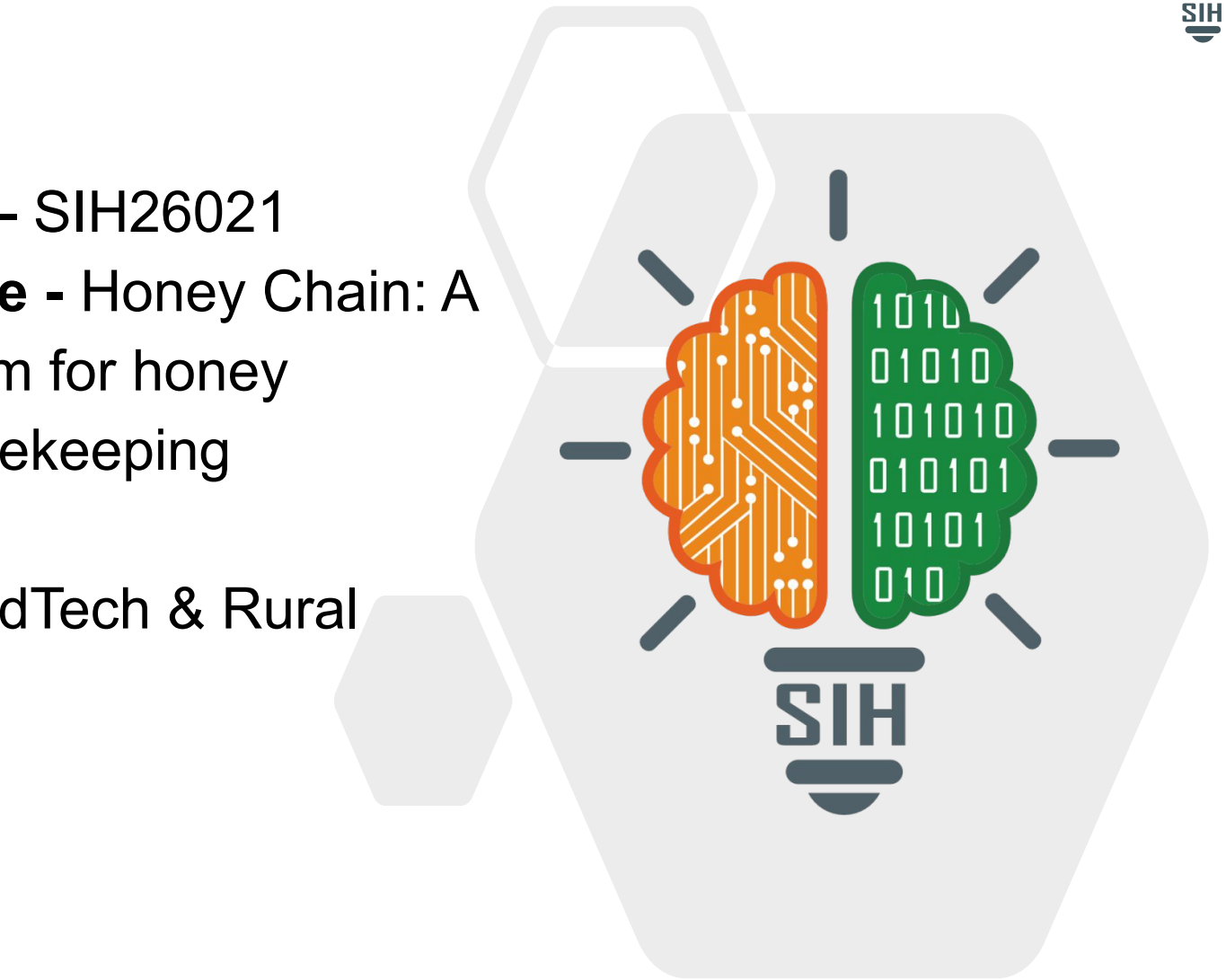
SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

- **Problem Statement ID –** SIH26021
- **Problem Statement Title -** Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category-** Software
- **Team ID-** 169191
- **Team Name -** KA1RO$



KA1RO$

HONEYCHAIN

Blockchain-Based Honey Traceability & Smart Beekeeping System

SMART INDIA HACKATHON 2026

Proposed Solution

HoneyChain is an end-to-end digital traceability system that connects beekeepers, laboratories, processing units, logistics, retailers and consumers on a single platform.

- **Harvest & Quality:** Digitally records harvest data and links verified lab reports to each batch.
- **Processing & Tracking:** Assigns every bottle a unique digital identity and records its movement across the supply chain.
- **Consumer Verification:** Enables consumers to scan the QR and verify the bottle's **origin, quality and ownership**.

Thus, HoneyChain creates a traceable journey from ***Hive*** → ***Consumer***, replacing fragmented records with a connected and verifiable digital chain.

How it Addresses the Problem :-

Stops Adulteration
Purity gates + volume locking

Prevents Counterfeits
Single-use QR + NFC verification

Secures Traceability
Blockchain records origin, custody and transactions

Builds Trust
Consumers can verify lab results and origin

Key Innovations / USPs

1. Mathematical Volume Lock
Harvest yield limits packaged volume.

2. Stateful QR + POS
Serialized bottle; SOLD after checkout; duplicate-scan fraud alert.

3. NFC Tamper-Evident Seal
Physical seal + digital verification.

4. Cryptographic Blockchain + IPFS
Immutable custody + lab records.

5. AI + IoT Smart Hive
Health and productivity insights.

6. Multilingual Consumer Support
Regional-language verification.

KA1RO$

TECHNICAL APPROACH

SMART INDIA HACKATHON 2026

HONEYCHAIN - End-to-End Process Flow

1 Smart Hive & IoT



- IoT sensors capture temperature, humidity & hive health (acoustic data)
- AI Analysis Dashboard

2 Beekeeper Harvest



- Logs raw yield (kg)
- Geolocation
- Botanical source

3 KVIC Accredited Lab



- NMR & purity tests Moisture ≤ 20%, C4 Sugar ≤ 7%
- Certificate generated
- Report hash → IPFS

4 Processing & Packaging



- Volume lock (Yield × 2)
- Unique serialized QR
- NFC seal (tamper-proof)

5 Secure Logistics



- Custody handover at each transit point via blockchain wallet signatures

6 Retail Store Arrival



- Shipment received
- Added to store inventory
- NFC seal (intact)

7 Store Inventory



- All bottles added to active stock
- Inventory tracking

8 At Billing Counter



- Customer Picks bottle for purchase
- Retail POS

9 Retailer Scans QR



- Pre-sale verification
- NFC scan

10 Status Check



Valid & In-Stock → Continue Sale
Already Sold / NFC Broken → Fraud Alert

11 Customer Details



- Enter Customer Details : Name, mobile Store ID

12 Blockchain Checkout



- Record sale permanently on blockchain ledger

13 e-Bill Generated



- Mark bottle as SOLD
- Reduce stock & generate KVIC digital e-bill

Tech Stack

Solidity 0.5.16, Ethereum EVM	Truffle Suite, Ganache	React.js	Web3.js 1.10	Node.js	IPFS	Keccak-256 Hashing	Tailwind CSS
Blockchain	Tooling & Test	Frontend	Web3	Backend / Runtime	Data Storage	Security & Integrity	Styling

14 Consumer Verification



- Scan QR / NFC tag
- View lab reports
- Origin & ownership proof

4-Tier Architectural Pipeline

01 DATA CAPTURE



- Node.js Digital Twin + Beekeeper Portal
- Hive telemetry + harvest weight + AI Analysis

02 QUALITY GATE



- KVIC Lab Spectrometry + IPFS
- Purity validation + report anchoring

03 CORE LEDGER



- HoneyChainCore.sol + Ethereum EVM
- Volume lock + custody + bottle registry

04 INTERFACE LAYER



- Retail POS + Consumer Passport
- QR verification + e-Bill + NFC Seal Scan

KA1RO$

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

FEASIBILITY

- **Integrated Platform** — Web + mobile, multilingual, role-based
- **IoT + AI** — Hive monitoring & insights
- **Secure Traceability** — QR, blockchain & IPFS

VIABILITY

- **Authenticity & Volume Control** — QR, NFC & volume locking
- **End-to-End Value** — Benefits all supply-chain stakeholders
- **Scalable Adoption** — Pilot-ready & expandable

PRACTICAL IMPLEMENTATION

- **Supply-Chain Tracking** — Harvest to retail
- **QR + NFC Verification** — Unique identity & tamper detection
- **Consumer Passport** — Provenance, authenticity & lab data

Potential Challenges & Risks

Challenge	Strategy
Weak/intermittent rural connectivity	**Offline-first sync** stores data locally and updates when connected.
Adoption by beekeepers & small actors	Simple role-based portals + multilingual support
Sensitive farm, lab & customer data	Encryption + access control + tamper-evident records.
Harsh field conditions for IoT	Robust IoT pipeline + suitable sensors + self-checks.
Data exchange across multiple actors	**Standardized APIs + shared IDs** for seamless synchronization.

Strategies For Overcoming Challenges

KA1RO$

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

1. ECONOMIC BENEFITS

- Protects honey value
- **Fairer prices for beekeepers**
- Reduces counterfeit & supply-chain losses
- Improves inventory & billing

2. SOCIAL BENEFITS

- Builds consumer trust
- **Empowers rural & tribal beekeepers**
- Enables origin & lab-result verification
- Strengthens KVIC clusters & market linkages

3. ENVIRONMENTAL BENEFITS

- Reduces supply-chain waste
- Prevents overproduction through volume locking
- Reduces paper dependency
- Supports sustainable hive monitoring

STAKEHOLDERS & IMPACTS



1. BEEKEEPER

Records harvest quantity and hive/origin information.

Better transparency & accountable farm-level records.

2. KVIC LAB

Verifies honey quality and certifies the batch.

Tamper-evident quality records & easier verification.

3. PROCESSOR / PACKAGING

Receives certified honey and packages permitted quantity.

Prevents undocumented volume expansion.

4. DISTRIBUTOR / LOGISTICS

Handles movement and custody handovers.

Traceable movement & reduced diversion risk.

5. RETAILER

Manages bottle inventory and verifies QR and NFC at sale.

Live inventory & duplicate/counterfeit detection.

6. CONSUMER

Scans the bottle QR and NFC seal to verify its record.

Greater transparency before and after purchase.

GOVERNMENT & REGULATORY BODIES

- Traceable digital records support audits and compliance.
- Immutable lab records reduce dependence on manual certificates.
- Supports KVIC National Honey Mission and rural livelihood goals.

KA1RO$

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

📖 Research & References

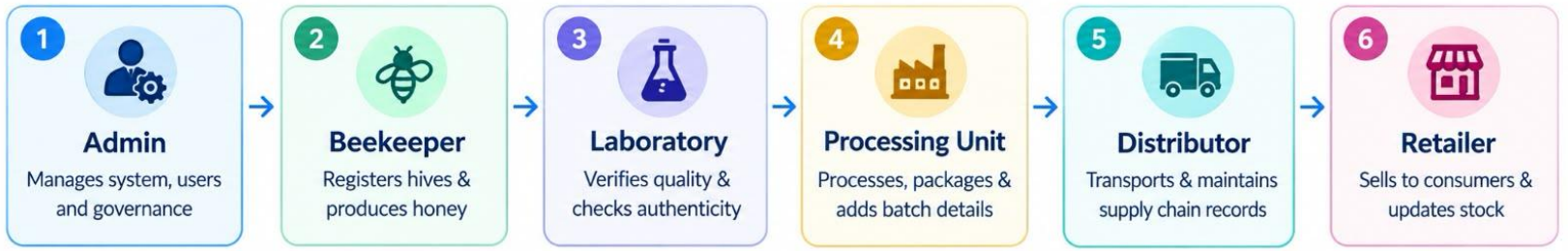
1. **FSSAI 2020 Honey Standards Direction** — Official operationalization of revised honey parameters including Specific Marker for Rice Syrup and Foreign Oligosaccharides. link

2. **CSE Investigation on Honey Adulteration** — Laboratory investigation demonstrating sophisticated honey adulteration and the use of C4/C3 sugar, rice-syrup markers and NMR testing. link

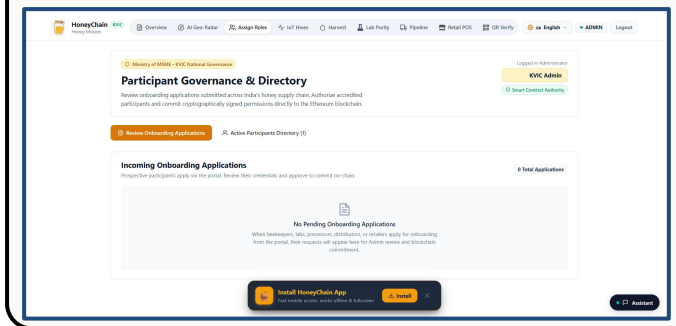
3. **KVIC National Honey Mission** — Government initiative supporting scientific beekeeping, honey production and rural beekeeper development. link

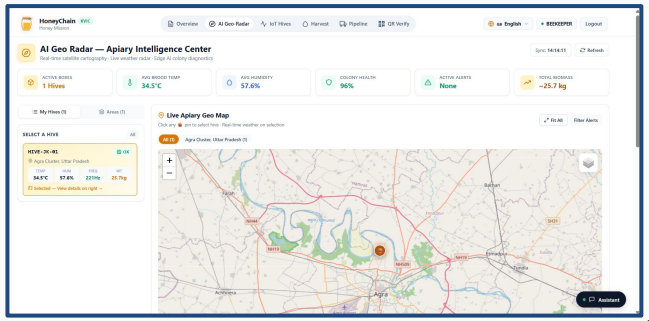
4. **Integrating Blockchain and IoT for Short Honey Supply Chains** — 2026 research directly addressing honey authenticity and quality through blockchain-IoT traceability and process modelling. link

👥 Platform Ecosystem - 6 Roles



🖥️ Prototype Screens





▶️ Demo

🔗 Prototype

📄 Report

📱 Consumer Verification Portal

QR Scan → Authenticity → Batch Information

